

## SECTION B

### 7 (short)

(a) Briefly explain why, for a given applied stress, the survival probability  $P_s$  of a brittle material reduces as the volume of material increases. [2]

(b) An axially constrained cylinder of length  $L$  has a solid circular cross section of outer radius  $R$ . Due to the manufacturing process, it develops a residual tensile stress that acts parallel to the axis of the cylinder, and varies linearly with the radial coordinate  $r$  ( $0 \leq r \leq R$ ) as follows

$$\sigma(r) = \frac{\alpha r}{R^4}$$

where  $\alpha$  is a constant. The cylinder is made from a brittle material. Using a suitable relationship for the Weibull statistics of fracture from the Materials Databook, derive an expression for the survival probability  $P_s$ . [4]

(c) Taking the Weibull modulus  $m = 3$  (as defined in the Materials Databook), calculate the value of  $P_s$  for a cylinder of radius  $R = 17$  mm, given that  $P_s = 0.9$  for a cylinder of radius  $R = 15$  mm.  $L$  is the same in each case. Comment on the result. [4]